2021 Ph H2 Q4

Section: Our Dynamic Universe

Topic: The Expanding Universe

A student analyses the light from a distant galaxy.

(a)
Calculate the recessional velocity of the galaxy

Use the **redshift formula**:

$$z = \frac{v}{c} \quad \Rightarrow \quad v = z \cdot c$$

Given:

- Redshift $z = 0.013$
- Speed of light $c = 3.0 \times 10^8 \, \mathrm{m\,s^{-1}}$

$$v = 0.013 \times 3.0 \times 10^8 = 3.9 \times 10^6 \, \mathrm{m\,s^{-1}}$$

(b)
Calculate the distance to the galaxy

Use **Hubble's Law**:

$$v = H_0 d \quad \Rightarrow \quad d = \frac{v}{H_0}$$

Given:

- $H_0 = 2.3 \times 10^{-18} \, \mathrm{s^{-1}}$
- $v = 3.9 \times 10^6 \, \mathrm{m\,s^{-1}}$

$$d = \frac{3.9 \times 10^6}{2.3 \times 10^{-18}} = 1.7 \times 10^{24} \, \mathrm{m}$$

(c)
Describe how this supports the theory of the Big Bang

✅ The light from distant galaxies is **redshifted**, meaning they are **moving away** from us.
✅ The further away the galaxy, the faster it recedes (Hubble's Law).
✅ This implies the Universe is **expanding**, which supports the idea that it originated from a **hot, dense state** — the **Big Bang**.

🧠 Revision Tips

- Redshift $z = \frac{v}{c}$ applies only for **low-speed galaxies** (not close to light speed)
- Use Hubble's Law $v = H_0 d$ to connect **velocity** and **distance** of galaxies
- Distance in metres! Always check units of H_0
- The more distant the galaxy, the greater the redshift — this is strong evidence for an **expanding universe**